

# Connecting EXHALE to the Lower Atmosphere: Survey, Implementation, and Example Applications

## 1. Motivation and current state

EXHALE's lower boundary is the  $\sim 1 \mu\text{bar}$  level: the input "Planet radius" is the *radius of that level* (user-guessed), the base temperature is fixed at  $T_0 = T_{\text{eq}}$ , the base density  $n_0$  is an input (or pressure-anchored via `base_p_ubar`), and the gas is purely atomic H/He + trace metals at prescribed abundances. No molecules exist anywhere in the domain. The Wind-AE IC port carries a *simplified* molecular layer ( $\mu$ -blend `molec_adjust` + bolometric heating/cooling, auto-shutoff when the atomic transition reaches the base).

This memo (i) surveys how the literature connects escape models to the lower atmosphere, (ii) documents the machinery implemented in `src/modules/lower_atmosphere/` and `src/utis/run_lower.py`, (iii) applies it to the four planets modelled with EXHALE (HD 209458 b, HD 189733 b, WASP-121 b, WASP-52 b), and (iv) prescribes how the outputs are used and connected to EXHALE runs.

## 2. What the literature does at the interface

| Approach                   | Lower atmosphere                                                                                                                                                            | Hands to the wind model at $\sim 1 \mu\text{bar}$                                                                                                                                 | Cost / availability                                                                 |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| Salz et al. (2016), TPCI   | none — CLOUDY solves the whole column (molecules disabled)                                                                                                                  | n/a (single domain; base at $n = 10^{14} \text{ cm}^{-3}$ , $T_{\text{eq}}$ )                                                                                                     | $\sim 3 \times 10^5$ CPU-h for 18 planets; interface not public                     |
| Koskinen et al. (2022)     | <i>analytic</i> : isothermal- $T_{\text{eq}}$ hypsometric column from 1 bar, $\mu(p)$ from chemical-equilibrium $\text{H}_2/\text{H}/\text{He}$ fits (Visscher et al. 2006) | base <i>radius</i> $r_0(1 \mu\text{bar})$ , $T_0=T_{\text{eq}}$ , $q_{\text{H}_2}/q_{\text{H}}/q_{\text{He}}$ ; base $v$ from mass conservation                                   | trivial (closed-form; all equations in the paper)                                   |
| Huang et al. (2017)        | <i>analytic spacer</i> : isothermal $\mu=2.3$ column, 1 bar $\rightarrow 10 \mu\text{bar}$                                                                                  | base radius of the atomic layer; $\text{Ly}\alpha$ absorbing bottom boundary ( $\text{H}_2$ accidental resonances, $\tau=1$ at $N_{\text{H}_2} \approx 10^{14} \text{ cm}^{-2}$ ) | trivial                                                                             |
| Taylor et al. (2025, 2026) | full Lavvas & Arfaux (2021) model (RC temperature + $\sim 150$ -species kinetics of Lavvas et al. 2014 + haze microphysics; $10^3 \rightarrow 10^{-6}$ bar)                 | $T(1 \mu\text{bar})$ , altitude of the $1 \mu\text{bar}$ level, composition (with a <i>tolerated discontinuity</i> ), $K_{\text{zz}}$                                             | in-house, <b>not public</b> ; coupling is a manual splice, iterated $\sim 3 \times$ |

Key quantitative facts:

- **Base insensitivity for strong winds** (Salz et al. 2016, §3.5–3.6):  $T_{\text{base}} \pm 40\% \Rightarrow \dot{M} \pm 5\text{--}10\%$  (linear); base density  $10^{14} \rightarrow 10^{12} \text{ cm}^{-3} \Rightarrow \dot{M} \lesssim 40\%$  lower. Their error budget: base density  $\pm 50\%$ , base  $T \pm 10\%$  — both dwarfed by the XUV uncertainty ( $\times 3$ ).  $\Rightarrow$  EXHALE’s fixed- $T_{\text{eq}}$  base is already defensible for hot Jupiters.
- **Where the fixed base breaks:** (i) planets near radiative equilibrium (stable thermospheres); (ii) cool/sub-Neptune planets where  $\text{H}_2$  survives into the wind — Salz’s GJ 1214 b test:  $\text{H}_2$  10–25% throughout,  $\text{H}^-$  a significant coolant,  $\dot{M} - 15\%$ ; Taylor et al. (2026): adding  $\text{H}_2$  halves the He 10830 signal (14% $\rightarrow$ 7%); Koskinen et al. (2022) hot Uranus: base  $q_{\text{H}_2} \approx 0.84$  and  $\text{H}_3^+$  IR emission is the dominant radiative coolant below  $\sim 2.3 R_p$ .
- **What a photochemical lower model uniquely provides** (Lavvas et al. 2014; Lavvas & Arfaux 2021): the  $\text{H}_2 \rightarrow \text{H}$  transition can sit exactly at 1  $\mu\text{bar}$  and is temperature-sensitive there; photochemistry (OH-catalysed  $\text{H}_2$  destruction) dissociates  $\text{H}_2$  even at 500–1000 K; *atomic-metal release profiles* (Na, K, Mg, Ca, Fe, Al, Si return to atomic form below  $\sim 10^{-3}$  bar) — the physically-motivated base abundances for EXHALE’s trace metals; photoionization-dominated electron density ( $\sim 10^8 \text{ cm}^{-3}$ ,  $100\times$  Saha); radical/haze heating of  $\pm 100\text{--}400$  K right at the handoff level.
- **The published coupling is loose, not monolithic:** Lavvas et al. (2014) take  $T(p < 1 \mu\text{bar})$  from the Koskinen thermosphere and use “species  $< 3$  amu escape at the wind velocity” as their upper BC; Taylor et al. tolerate a composition discontinuity at 1  $\mu\text{bar}$ . Only TPCI runs a tight per-timestep coupling (and it disables molecules).

### 3. The three-tier plan

**Tier 1** — analytic lower column (Koskinen-2022 style): derive the 1- $\mu\text{bar}$  base radius and base  $\text{H}_2/\text{H}/\text{He}$  from a closed-form column. **Tier 2** — molecular extension of EXHALE ( $\text{H}_2$ ,  $\text{H}_2^+$ ,  $\text{H}_3^+$ ,  $\text{HeH}^+$  chemistry +  $\text{H}_3^+$  IR cooling), required only for warm Neptunes / sub-Neptunes; **Tier 2a** — an interim EOS-only molecular-base correction. **Tier 3** — consume a real lower-atmosphere model (photochemistry + radiative–convective  $T(p)$ ) through a file interface; the open-source VULCAN (Tsai et al. 2017, 2021) + HELIOS (Malik et al. 2017) stack is the practical replacement for the non-public Lavvas & Arfaux code, and a TPCI-style full coupling is rejected (cost, availability, and demonstrated base insensitivity). Which science case needs which tier: hot Jupiters  $\rightarrow$  Tier 1 only; He 10830/ $\text{H}\alpha$  work  $\rightarrow$  Tier 1; metal transmission lines  $\rightarrow$  Tier 3 (atomic-metal release abundances); warm Neptunes / sub-Neptunes  $\rightarrow$  Tier 2 + 1.

### 4. Implementation

All Fortran lives in `src/modules/lower_atmosphere/`; the Python driver in `src/utils/`. Everything is **opt-in / default off** — with no keys and no `base.inp`, standard runs are unchanged (regression: HD 209458 b He  $2^3\text{S}$ +metals 3000-step  $\log \dot{M} = 8.97$ , identical to EXHALE\_v1.0).

#### 4.1 Tier 1: lower\_column.f90

**Equations.** The isothermal hypsometric relation is integrated from the 1-bar level (radius  $R_{1\text{bar}}$ , in practice the broadband transit radius) to the base pressure  $p_0 = 1 \mu\text{bar}$ ,

$$\frac{d \ln p}{dr} = -\frac{\mu(p, T) m_{\text{H}} g(r)}{k_{\text{B}} T}, \quad g(r) = \frac{GM_{\text{p}}}{r^2}, \quad T = T_{\text{eq}}, \quad (1)$$

with a fixed-step 4th-order Runge–Kutta in  $x = \ln p$  (2000 steps; the integrand is smooth). The mean molecular weight follows the chemical-equilibrium  $\text{H}_2/\text{H}$  partition of Visscher et al. (2006) as fitted by Koskinen et al. (2022, Eq. 11),

$$q_{\text{H}_2} \simeq \frac{1.9845 + 10^u - \sqrt{10^u (3.9690 + 10^u)}}{2.3670}, \quad u = -\frac{23672}{T} - \log_{10} p_{[\text{bar}]} + 6.2645. \quad (2)$$

An implementation subtlety uncovered during validation: Eq. (2) returns the *full-mixture* volume mixing ratio  $q_{\text{H}_2} = n_{\text{H}_2}/(n_{\text{H}_2} + n_{\text{H}} + n_{\text{He}})$  (a protosolar He abundance is baked into the fit), so the fraction  $x_2$  of H *nuclei* bound in  $\text{H}_2$  must be recovered as

$$x_2 = \frac{2q_{\text{H}_2}(1 + f_{\text{He}})}{1 + q_{\text{H}_2}} \leq 1, \quad \mu = \frac{1 + 4f_{\text{He}}}{(1 - x_2) + x_2/2 + f_{\text{He}}} \text{ [amu]}, \quad (3)$$

with  $f_{\text{He}} = n_{\text{He}}/n_{\text{H(nuclei)}}$  the elemental ratio (EXHALE’s HeH). Equation (3) was verified against the Koskinen et al. (2022) Model A base composition ( $q_{\text{H}_2} = 0.84$ ,  $q_{\text{H}} = 0.026 \Rightarrow x_2 = 0.985$ ); the naive H-subsystem inversion  $x_2 = 2q/(1 + q)$  is *wrong* and misses the gate.

**Interface.** `lower_column_solve(Mp, R_deep, T, fhe, p_deep, p_base, r_base, qh2_b, qh_b, qhe_b, mu_b, r_base_atomic)` (cgs; pressures in bar). Because chemical equilibrium *underestimates* H dissociation at  $T_{\text{eq}} \approx 1000\text{--}2000 \text{ K}$  (photochemistry ignored; Moses et al. 2011; Koskinen et al. 2013a), the routine also integrates the same column with the fully atomic  $\mu = (1 + 4f_{\text{He}})/(1 + f_{\text{He}})$  and returns `r_base_atomic`: the two radii *bracket* the truth. The public helper `q_h2_equilibrium(p, T)` exposes Eq. (2) (NaN-safe, clamped to  $[0, 1]$ ).

**EXHALE key.** Lower column: `<R_1bar in R_J>` in `input.inp` triggers a startup report: the derived  $r_0(1 \mu\text{bar})$  (equilibrium + atomic bracket), the input “Planet radius” for comparison, and the base  $q_{\text{H}_2}/q_{\text{H}}/q_{\text{He}}/\mu$ , plus a warning when the equilibrium base is strongly molecular. *Report only — nothing is overridden* (overriding is Tier 3’s job, §4.4).

**Validation gate (passed).** Koskinen et al. (2022) Model A (Uranus-mass planet,  $M = 8.6813 \times 10^{28} \text{ g}$ ,  $R_{1\text{bar}} = 2.5559 \times 10^9 \text{ cm}$ , Sun-like star at 0.05 au,  $T_{\text{eq}} = 1140 \text{ K}$ ,  $f_{\text{He}} = 0.0793$ ):

|                        | $r_0/R_{1\text{bar}}$ | $q_{\text{H}_2}(\text{base})$ | $q_{\text{H}}(\text{base})$ |
|------------------------|-----------------------|-------------------------------|-----------------------------|
| this module            | 1.343                 | 0.838                         | 0.027                       |
| Koskinen et al. (2022) | 1.34                  | $\approx 0.84$                | $\approx 0.026$             |

Approximations (documented in the source): isothermal column at  $T_{\text{eq}}$ ; spherical gravity (the Roche-equipotential correction matters only near RLO); chemical equilibrium only.

## 4.2 Tier 2a: the Molecular base key

Molecular base: True applies an *EOS-only* correction at startup: the H nuclei bound into H<sub>2</sub> at (1  $\mu$ bar,  $T_0$ ) per Eq. (2)–(3) are removed from the base particle budget,

$$\text{ntot\_bc} \rightarrow \text{ntot\_bc} - \frac{x_2/2}{1 + f_{\text{He}}}, \quad (4)$$

which lowers the base pressure ( $p_{\text{base}} = \text{ntot\_bc} n_0 k_B T_0$ ) and raises the base mean molecular weight, i.e. a heavier, more compact base — the leading-order effect of a molecular base on the hydrodynamics. The species arrays remain atomic (H<sub>2</sub> chemistry is Tier-2 proper), so this is a *crude, clearly-flagged* approximation. Example: HD 209458 b,  $q_{\text{H}_2}(1 \mu\text{bar}, 1450 \text{ K}) = 0.831 \Rightarrow \text{ntot\_bc} 1.0 \rightarrow 0.546$ .

## 4.3 Tier 2 foundation: h3p\_cooling.f90 and mol\_rates.f90

**H<sub>3</sub><sup>+</sup> infrared cooling.** Optically thin cooling per molecule from Miller et al. (2013, Table 5,  $z(T)$  fits, archived as references/Miller\_2013\_JPCA\_117\_9770.pdf):

$$\ln E(\text{H}_3^+, T) = \sum_n C_n T^n \quad [E \text{ in W molecule}^{-1} \text{ sr}^{-1}], \quad (5)$$

piecewise over 30–300, 300–800, 800–1800, and 1800–5000 K (fit errors < 0.1% over 300–5000 K), held frozen above 5000 K (H<sub>3</sub><sup>+</sup> is destroyed well below). The volumetric rate is

$$\Lambda_{\text{H}_3^+} = n_{\text{H}_3^+} 4\pi E(T) s_{\text{nonLTE}}(T, n_{\text{H}_2}) \times 10^7 \quad [\text{erg s}^{-1} \text{ cm}^{-3}], \quad (6)$$

where  $s_{\text{nonLTE}}$  is the departure factor of Miller et al. (2013, Table 6; their Oka & Epp detailed-balance approach), implemented as a bilinear interpolation in  $(T, \log_{10} n_{\text{H}_2})$  over  $T = 300\text{--}5000 \text{ K}$  and  $n_{\text{H}_2} = 10^6\text{--}10^{14} \text{ cm}^{-3}$ ;  $s \rightarrow 1$  (LTE) at high density, and the tabulated values are upper limits per the paper’s caveat on the proton-hopping rate. Both interpolators use the NaN-safe .not. (x>edge) bracketing adopted after the 2026-07-02 code review. **Validation:** the fits reproduce the paper’s Table-4 anchors at  $T = 500/1000/2000/3000/5000 \text{ K}$  to <0.5%, and  $s(1000 \text{ K}, 10^{10} \text{ cm}^{-3}) = 0.4955$ ,  $s(2000 \text{ K}, 10^{12}) = 0.9499$  exactly.

**Molecular reaction rates.** mol\_rates.f90 provides the full H<sub>2</sub>/H<sub>2</sub><sup>+</sup>/H<sub>3</sub><sup>+</sup>/HeH<sup>+</sup> thermochemical network of Koskinen et al. (2022, Table 1; transcribed and verified against the PDF page): recombination (R1–R2, Storey & Hummer 1995), Voronov (1997) collisional ionization (R3–R4), H<sub>2</sub><sup>+</sup>/H<sub>3</sub><sup>+</sup>/HeH<sup>+</sup> dissociative recombination (R5–R7, R16; Larsson et al. 2008), the H<sub>3</sub><sup>+</sup> production backbone H<sub>2</sub><sup>+</sup> + H<sub>2</sub> → H<sub>3</sub><sup>+</sup> + H (R8,  $2 \times 10^{-9}$ ), thermal and electron-impact H<sub>2</sub> dissociation (R12, R14), three-body chemistry (R13, R15), the HeH<sup>+</sup> chain (R17–R20), and H/He charge exchange (R21–R23). The stellar photo-rates P1–P5 (including H<sub>2</sub> dissociative and double photoionization) follow EXHALE’s existing PH\_heat machinery once H<sub>2</sub> cross sections are added. Baseline caveat inherited from the source: Lyman–Werner neutral photodissociation is not in the network (Koskinen et al.’s sensitivity test bounded its effect on  $\dot{M}$  at  $\times 1.4$ ).

**Remaining Tier-2 work (open).** The coupled System\_HeH\_mol ionization system (four new unknowns, following the merged System\_HeH\_TR\_metals pattern), the EOS extension ( $\mu$  and  $n_{\text{tot}}$  with H<sub>2</sub> in composition.f90, diatomic dof, the 4.48 eV dissociation sink), and H<sub>2</sub> photo cross sections. Gates: a hot Jupiter with molecules on must reproduce the atomic code; a GJ 1214 b-like case must halve the He 10830 line (Taylor et al. 2026) with  $\dot{M}$  shifts  $\lesssim 40\%$ .

#### 4.4 Tier 3: base.inp and run\_lower.py

**The handoff file.** An optional base.inp in the run directory is read by input\_read *before* the derived constants, so its overrides propagate consistently ( $v_0$ ,  $b_0$ , grid, normalization):

| key      | unit               | overrides                                 | used by                   |
|----------|--------------------|-------------------------------------------|---------------------------|
| T_base   | K                  | $T_0$ (base temperature)                  | thermal BC, EOS scales    |
| r_base   | $R_J$              | “Planet radius” ( $= r_0$ of 1 $\mu$ bar) | grid, gravity, areas      |
| HeH_base | –                  | He/H number ratio                         | composition, He diffusion |
| Kzz_base | cm <sup>2</sup> /s | he_kzz                                    | He_diffusion homopause    |

Every override is echoed to stdout; an absent file is a strict no-op. Unknown keys and # comments are ignored, so a richer lower-atmosphere model may carry extra metadata in the same file.

**The driver.** src/utils/run\_lower.py <run\_dir> --r1bar <R\_J> [--guillot] [--tint T] [--kzz K] reads  $M_p$ ,  $T_{eq}$ , He/H from the run directory’s input.inp, integrates the same column as §4.1 in Python, and writes base.inp. Two temperature models: isothermal  $T_{eq}$  (default; identical to the Fortran Tier-1 column) and the Guillot (2010) semi-grey global-average profile

$$T^4(\tau) = \frac{3T_{int}^4}{4} \left( \frac{2}{3} + \tau \right) + \frac{3T_{eq}^4}{4} \left( \frac{2}{3} + \frac{1}{\gamma\sqrt{3}} + \left[ \frac{\gamma}{\sqrt{3}} - \frac{1}{\gamma\sqrt{3}} \right] e^{-\gamma\tau\sqrt{3}} \right), \quad \tau = \kappa_{IR} p/g, \quad (7)$$

with standard hot-Jupiter defaults ( $\kappa_{IR} = 10^{-2} \text{ cm}^2 \text{ g}^{-1}$ ,  $\gamma = 0.4$ ,  $T_{int} = 100 \text{ K}$ ). The file records the base  $q_{H_2}/q_H/q_{He}$  as comments and prints a molecular-base warning when  $q_{H_2} > 0.3$ . The upgrade path — running VULCAN for the composition (including the photochemical  $H_2$  dissociation and the atomic-metal release that feed a metals.inp) and a real RC model for  $T(p)$  — writes the *same* base.inp keys, so nothing on the EXHALE side changes.

## 5. Example applications: HD and WASP planets

The machinery was applied to the four planets modelled with EXHALE (examples/13\_lower\_atmosphere/;  $R_{1bar}$  = broadband transit radius;  $T_{eq}$ ,  $M_p$ , He/H from each run’s input.inp).

|             | $M_p$   | $T_{eq}$ | He/H   | $R_{1bar}$ | input $R_0$ | $r_0(1\ \mu\text{bar})\ [R_J]$ | $r_0^{\text{Gui}}$ | $q_{\text{H}_2}$ | $\mu_{\text{base}}$ |       |
|-------------|---------|----------|--------|------------|-------------|--------------------------------|--------------------|------------------|---------------------|-------|
| planet      | $[M_J]$ | [K]      |        | $[R_J]$    | $[R_J]$     | chem.eq.                       | atomic             | $[R_J]$          | (base)              | [amu] |
| HD 209458 b | 0.720   | 1450     | 0.0833 | 1.36       | 1.401       | 1.472                          | 1.582              | 1.466            | 0.831               | 2.25  |
| HD 189733 b | 1.237   | 1183     | 0.0833 | 1.138      | 1.193       | 1.174                          | 1.205              | 1.171            | 0.838               | 2.26  |
| WASP-121 b  | 1.182   | 2358     | 0.0851 | 1.753      | 2.2075      | 2.010                          | 2.133              | 1.966            | 0.005               | 1.24  |
| WASP-52 b   | 0.460   | 1304     | 0.0204 | 1.27       | 1.270       | 1.437                          | 1.615              | 1.428            | 0.837               | 1.95  |

Isothermal- $T_{eq}$  column (chem. equilibrium + fully-atomic bracket) and the Guillot variant  $r_0^{\text{Gui}}$  ( $T_{base}^{\text{Gui}} = 1313, 1072, 2136, 1181 \text{ K}$  respectively). “input  $R_0$ ” is the base radius currently used in the production runs.

Interpretation (per planet):

- **HD 189733 b — consistent.** The production base radius (1.193  $R_J$ ) falls *inside* the derived bracket [1.174, 1.205]: the adopted setup is validated by the column, not merely assumed.

- **HD 209458 b — input  $\sim 5\%$  low.** The production value  $1.401 R_J$  sits just below the bracket  $[1.472, 1.582]$ . Given the isothermal- $T_{\text{eq}}$  assumption this is within the method’s spread, but it suggests the  $1\text{-}\mu\text{bar}$  level may lie slightly higher than assumed; the effect on  $\dot{M}$  is bounded by the Salz base-sensitivity results ( $\lesssim 10\%$ -level).
- **WASP-121 b — the atomic base is *verified*.** At  $T_{\text{eq}} = 2358 \text{ K}$  the equilibrium base is already atomic ( $q_{\text{H}_2} = 0.005$ ,  $q_{\text{H}} = 0.92$ ,  $\mu = 1.24$ ): for this ultra-hot Jupiter EXHALE’s atomic assumption holds *even in equilibrium*, with no appeal to photochemistry. The production base radius  $2.2075 R_J$  (derived earlier from the Huang et al. 2023 Table-3  $\log g$  anchoring) lies  $\sim 4\text{--}10\%$  above the isothermal- $T_{\text{eq}}$  bracket  $[2.010, 2.133]$  — consistent with a dayside lower atmosphere hotter than the redistribution-averaged  $T_{\text{eq}}$ , which extends the column. Note the deep layers are still molecular ( $q_{\text{H}_2}(1 \text{ bar}) \approx 0.83$ ), so  $\mu$  varies from 2.25 to 1.24 along this column — the equilibrium track is *shorter* than the atomic bracket here because of its heavier deep-layer  $\mu$ .
- **WASP-52 b — misplaced base, impact quantified.** The production runs use the transit radius itself ( $1.27 R_J$ ) as the  $1\text{-}\mu\text{bar}$  base — the common shortcut — but the column places that level at  $1.44\text{--}1.62 R_J$ , i.e. the base is  $0.17\text{--}0.35 R_J$  ( $13\text{--}27\%$ ) too deep. *Recomputation* (the `fxuv1p0_he98` configuration rerun Newton-converged with the corrected equilibrium-column base  $r_0 = 1.437 R_J$ , all else identical):  $\log \dot{M} 11.97 \rightarrow 12.14$  ( $\times 1.5$ ), but the He 10830 observable barely moves — line-center absorption  $27.11 \rightarrow 27.56\%$  ( $\times 1.02$ ) and equivalent width  $0.713 \rightarrow 0.722 \text{ \AA}$  ( $\times 1.01$ ). The line-forming effective radius is set by the  $\tau=1$  surface of the extended wind, which stays nearly fixed in *physical* units (the larger, less-bound base raises  $\dot{M}$ , but the He  $2^3\text{S}$  column at a given physical radius is almost unchanged: peak  $n_{2^3\text{S}} 150.5 \rightarrow 147.6 \text{ cm}^{-3}$ ). Conclusion: the base misplacement biases the *mass-loss rate* at the  $\times 1.5$  level but appears not to bias the He 10830 scan conclusions materially. Also note this run family’s low He/H ( $= 0.0204$ , the “He98” abundance scan), visible in the light  $\mu_{\text{base}}$ .
- **Common pattern.** All three  $T_{\text{eq}} \lesssim 1500 \text{ K}$  planets have strongly molecular *equilibrium* bases ( $q_{\text{H}_2} \approx 0.83$ , the fully-molecular limit of the fit) — quantifying that their atomic bases rest on *photochemical* dissociation (Moses et al. 2011; Koskinen et al. 2013a) rather than thermal, which is precisely what the Tier-3 VULCAN upgrade would supply from first principles. The Guillot column is systematically slightly more compact (cooler skin above the photosphere) — a  $\lesssim 0.5\%$  radius effect at fixed assumptions.

## 6. Using and connecting the results

### 6.1 Recommended per-planet workflow

1. **Column check (always, free).** Add `Lower column: <transit radius>` to `input.inp` (or run `run_lower.py`). Read off: does the current “Planet radius” fall inside the `[chem.eq., atomic]` bracket? Is  $q_{\text{H}_2}(\text{base})$  small?
2. **Decide the tier.**  $q_{\text{H}_2} \lesssim 0.1$  (e.g. WASP-121 b): the atomic model is verified — proceed as usual.  $q_{\text{H}_2} \sim 0.8$  with  $T_{\text{eq}} \sim 1000\text{--}2000 \text{ K}$  (typical hot Jupiters): the atomic base rests on photochemistry — proceed, but note the assumption; a genuinely cool planet (warm Neptune / sub-Neptune) needs Tier-2 physics before its He 10830 predictions are trusted.

3. **Adopt the geometry.** If the input radius lies outside the bracket (WASP-52 b), either correct “Planet radius” by hand or copy the driver’s `base.inp` into the run directory so EXHALE adopts  $r_0$ ,  $T_0$ , He/H and  $K_{zz}$  automatically (echoed at startup).
4. **Run EXHALE** with the usual two-stage convergence (+ Newton finish for quantitative  $\dot{M}$ ), then `TPM.py` for the transit spectra.
5. **(Optional) iterate** once a real lower-atmosphere model is attached (§6.3).

## 6.2 Where each quantity connects

| quantity                                                      | produced by                          | consumed by / effect                                                                                                                               |
|---------------------------------------------------------------|--------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| $r_0$ (1 $\mu$ bar)                                           | Tier 1 / <code>base.inp</code>       | grid inner radius, gravity $b_0$ , irradiated area; TPM continuum radius                                                                           |
| bracket [ $r_0^{\text{eq}}, r_0^{\text{atom}}$ ]              | Tier 1                               | consistency check on the input radius; systematic-error bar on $r_0$                                                                               |
| $T_{\text{base}}$                                             | driver ( $T_{\text{eq}}$ or Guillot) | thermal BC $T_0$ ; $v_0$ , $p_0$ , $b_0$ normalization                                                                                             |
| $q_{\text{H}_2}$ (base)                                       | Eq. (2)                              | tier-decision switch; Molecular base correction; Tier-2 initial condition                                                                          |
| He/H (base)                                                   | <code>base.inp</code>                | composition; the He_diffusion reservoir value                                                                                                      |
| $K_{zz}$                                                      | <code>base.inp</code>                | he_kzz: homopause of the He/H (and metal) diffusive separation                                                                                     |
| $\Lambda_{\text{H}_3^+}(T, n_{\text{H}_3^+}, n_{\text{H}_2})$ | <code>h3p_cooling</code>             | (Tier 2) new term in the cool array of the energy equation                                                                                         |
| R1–R23 rates                                                  | <code>mol_rates</code>               | (Tier 2) source terms of <code>System_HeH_mol</code> ; $\text{H}_3^+/\text{HeH}^+$ densities for the cooling and for IR-diagnostic post-processing |
| metal release $n_X/n_{\text{H}}$                              | (Tier 3, VULCAN)                     | <code>metals.inp</code> values — replaces assumed solar totals for the Mg II/Ca II/Na I transmission lines                                         |

## 6.3 Feedback loop (future, with a real lower model)

One-way coupling (lower  $\rightarrow$  `base.inp`  $\rightarrow$  EXHALE, with the Taylor-style tolerated composition discontinuity) is the default. For the two-way iteration: EXHALE already computes the attenuated stellar spectrum on its energy grid, so the *transmitted XUV flux at the base*,  $F_\nu(r_0) = F_\nu e^{-\tau_\nu(r_0)}$ , can be dumped and handed to the photochemical model as its top irradiation; and EXHALE’s  $\dot{M}$  supplies the lower model’s upper BC (“species  $< 3$  amu escape at the wind velocity”, exactly the Lavvas et al. 2014 prescription). Taylor et al. found  $\sim 3$  iterations sufficient for the analogous Ly $\alpha$  loop; the same is expected here since the wind is insensitive to base details at the  $\lesssim 10\%$  level (Salz et al. 2016).

## 6.4 Consistency cautions

- Do not double-count composition: if `base.inp` sets He/H from a lower model, the `metals.inp` abundances should be the same model’s *atomic release* values, not solar totals.
- `base.inp` overrides win over `input.inp` for  $T_0$ ,  $R_0$ , He/H (echoed); remove the file to return to the manual values.
- The Tier-1 column is isothermal and spherical: near-RLO planets (WASP-121 b) need the Roche correction before the bracket is taken quantitatively, and dayside-hot lower atmospheres extend beyond the  $T_{\text{eq}}$  column

(both visible in the WASP-121 b example above).

- The Molecular base key changes only the EOS budget; do not combine it with quantitative He 10830 claims for genuinely cool planets — that is Tier-2 physics.

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